DIGITAL TWIN · CARDIOLOGY

A Patient-Specific Digital Twin for Early Cardiac Congestion Prediction

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Happiest Health Digital Twin Challenge 2026 · Phase 1



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01 · CLINICAL PROBLEM

Fluid build-up in heart failure can go unseen

01	02	03	04
Heart weakens	Pressure backs up	Fluid in lungs	Emergency care
Heart failure lets fluid back up into the lungs and other parts of the body.	Current monitoring may not catch the change in time.	The first 30 days after heart treatment are the most critical.	If missed, the patient can become critical and need emergency treatment.

- Heart failure lets fluid back up into the lungs and other parts of the body.
- If missed, the patient can become critical and need emergency treatment.
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02 · THE MONITORING GAP

Once the patient goes home, the hospital loses sight

CENTRAL SPECIALTY CARE

City hospital

Specialized cardiologists, advanced diagnostics, and surgical intervention.

BLIND SPOT

Distant recovery

REMOTE POST-DISCHARGE

Home town

Patients recover far from specialist centers where regular check-ups are rare.

- Many towns have no heart specialist nearby, so regular check-ups are rare.
- After a heart attack or surgery, patients often recover at home in their home towns.
- The treating hospital cannot easily see how the heart is doing.
- Our target: heart patients far from a specialist, in the first 30 days.

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03 · SYSTEM OVERVIEW

A digital twin of the heart, fed by a chest patch

1	2	3
1. Chest patch ECG, chest movement, skin temperature	2. Model Estimates lung back-pressure and the chance of congestion	3. Digital twin and doctor dashboard A 3D heart that beats with the patient, a patient list, and a medicine what-if simulator

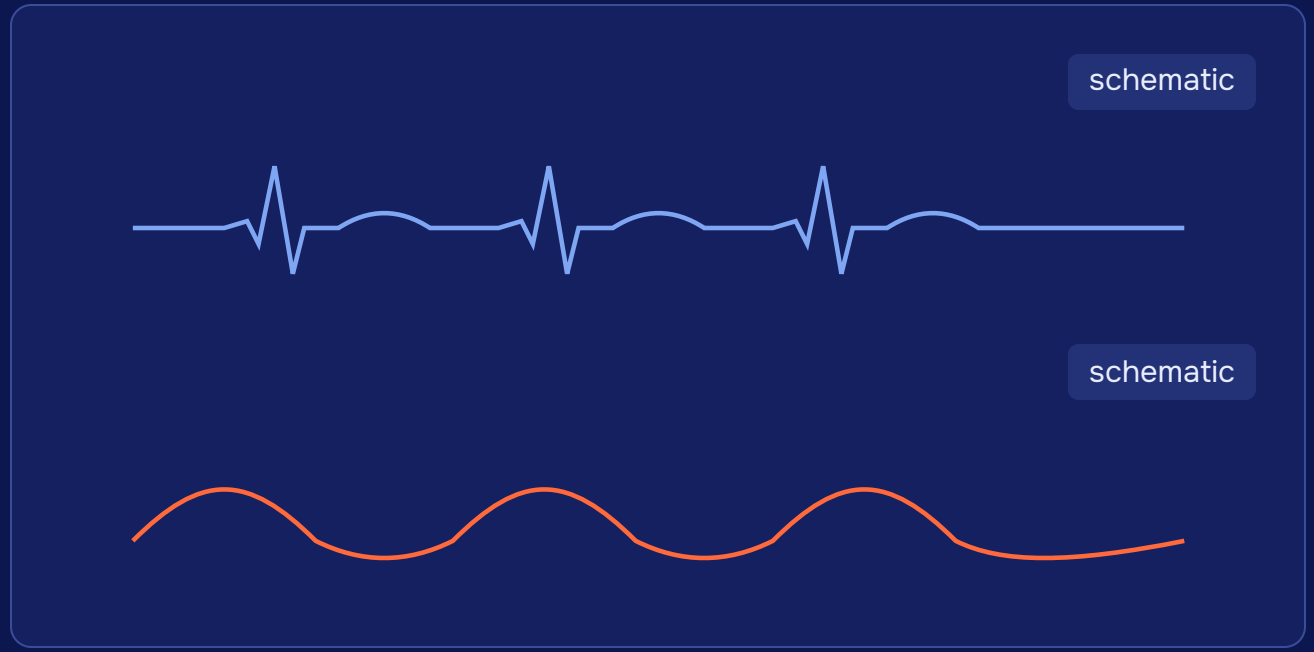
1. **Chest patch:** ECG, chest movement, skin temperature
2. **Model:** estimates lung back-pressure and the chance of congestion
3. **Digital twin and doctor dashboard:** a 3D heart that beats with the patient, a patient list, and a medicine what-if simulator

Lung back-pressure = pulmonary wedge pressure. Congestion = above 18 mmHg (CSI 2018 guideline).

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Real recordings from 73 heart failure patients

- Open PhysioNet SCG-RHC dataset: 83 recordings, 73 patients
- Chest patch: ECG, a 3-direction motion sensor, skin temperature
- Catheter: a thin tube placed inside the heart gives the true pressure
- Training: 63 clean recordings from 59 patients



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A support vector model: 20 inputs, two outputs

Inputs from the chest patch (15): heart rate, beat steadiness, heart vibration per beat, breathing rate and depth, body tilt, skin temperature

Inputs from the patient record (5): age, sex, BMI, two blood pressure values



Support Vector Regression

Outputs: (1) estimated lung back-pressure in mmHg, with an error range; (2) chance of congestion in %

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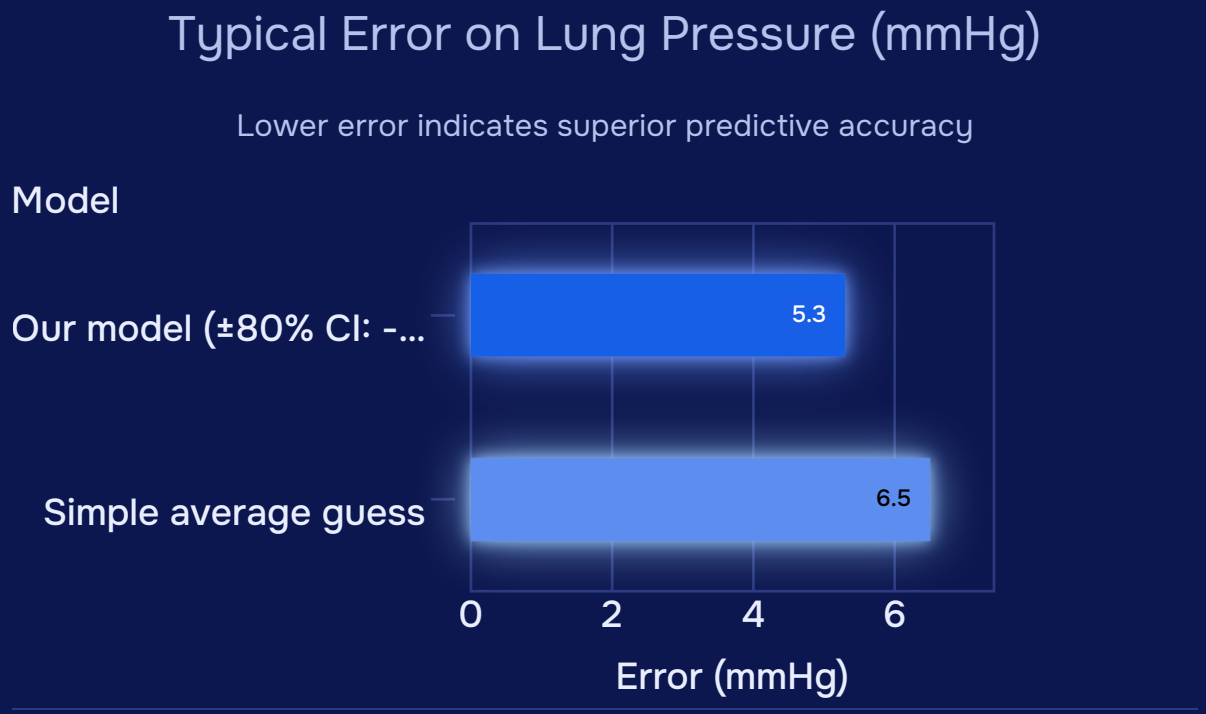
06 · EVALUATION

It beats a simple average guess on lung pressure

Tested on patients the model never saw. It gives the congested patient the higher chance about 3 times in 4.

59 patients is small, so the model is not precise for one person. Stroke volume, cardiac output and right-side pressures did no better than a guess, so we do not use them for decisions.

Measure	Our model	Simple average guess
Typical error, lung pressure	5.3 mmHg	6.5 mmHg
Error range, 8 in 10 patients	about -8 to +9 mmHg	not measured
Congestion ranking (AUROC)	0.75	0.5 = guessing



Error range (80% confidence interval): -8 to +9 mmHg on unseen test fold

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The doctor dashboard

Shown as a simple timeline on a title card for the live demo.

Patient list. Select one patient.

Patient tab: lung back-pressure with error range, chance of congestion, status, measured values.

Heart twin: the 3D heart beats on this patient's real beat timing. Red pulse = chance of 50% or more.

Medicine simulator: change a dose. The estimate, chance and heart update. Labelled "simulation".

Model and evidence tab: how good the estimate is.

Recommendation card and its safety flag.

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